

BERLIN SCHONEFELD, Germany

WMO: 103850

Lat: 52.3808N

Lon: 13.5306E

Elev: 165

StdP: 14.61

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.8	(c) 17.2	(d) 5.1	(e) 7.2	(f) 15.2	(g) 10.3	(h) 9.4	(i) 19.0	(j) 29.9	(k) 43.5	(l) 27.4	(m) 43.3	(n) 7.5	(o) 60	(p) 0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.2	(c) 88.1	(d) 66.9	(e) 84.5	(f) 65.7	(g) 80.9	(h) 64.9	(i) 69.6	(j) 81.5	(k) 68.2	(l) 79.3	(m) 66.8	(n) 77.4	(o) 9.3	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	97.1	72.8	64.5	91.4	71.6	62.8	86.0	70.6	33.6	81.2	32.4	79.8	31.3	77.2	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 23.8	(b) 21.0	(c) 18.6	(e) 7.1	(f) 94.7	(g) 8.0	(h) 3.9	(i) 1.3	(j) 97.5	(k) -3.4	(l) 99.8	(m) -7.9	(n) 101.9	(o) -13.7	(p) 104.7		
			DB	7.1	94.7	8.0	3.9	1.3	97.5	-3.4	99.8	-7.9	101.9	-13.7	104.7	
			WB	6.5	73.1	8.0	1.4	0.8	74.1	-3.9	74.9	-8.4	75.7	-14.3	76.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	32.5	34.0	40.6	50.7	57.4	64.2	68.1	66.3	59.6	50.4	42.2	36.5
	DBStd	14.22	9.05	7.51	7.73	7.24	7.33	6.46	6.04	5.78	5.70	6.36	6.58	7.82
	HDD50	2121	541	448	299	79	15	0	0	0	1	75	245	418
	HDD65	5693	1006	868	756	430	255	89	37	51	180	453	685	883
	CDD50	2236	0	0	8	99	245	427	561	506	291	88	10	1
	CDD65	331	0	0	0	1	20	66	133	92	19	0	0	0
	CDH74	3165	0	0	0	23	235	678	1247	807	171	4	0	0
	CDH80	1005	0	0	0	2	57	212	430	275	29	0	0	0
(14) Wind	WSAvg	9.1	10.6	9.8	9.9	9.0	8.8	8.5	8.3	7.8	8.2	8.5	9.2	10.4
(15) Precipitation	PrecAvg	23.1	1.9	1.5	1.6	1.2	2.2	2.3	3.0	2.2	2.0	1.6	1.8	1.9
	PrecMax	30.9	3.6	3.3	3.8	3.1	4.8	6.5	8.1	4.8	5.3	3.9	3.9	4.1
	PrecMin	16.3	0.1	0.2	0.4	0.1	0.2	0.3	0.8	0.9	0.6	0.3	0.0	0.5
	PrecStd	3.7	0.9	0.7	0.9	0.6	1.2	1.3	1.6	1.0	1.1	0.9	0.8	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	53.8	66.2	77.1	86.1	91.1	94.7	93.2	84.3	73.2	60.7	55.0
		MCWB	49.3	45.2	52.6	59.1	64.9	67.7	67.4	67.8	65.6	61.6	56.5	50.9
	2%	DB	50.1	50.1	60.6	72.0	80.4	85.9	88.2	87.6	78.8	67.7	57.1	51.6
		MCWB	47.2	45.9	49.8	56.1	63.1	66.2	66.4	67.8	63.5	58.7	53.2	48.7
	5%	DB	46.8	47.8	55.7	68.3	75.4	82.1	84.5	82.2	75.0	64.2	53.8	49.6
		MCWB	44.5	43.9	47.6	54.8	60.3	65.4	66.0	65.7	62.0	57.2	50.2	46.9
	10%	DB	44.6	44.6	52.0	64.6	71.7	77.2	80.8	77.4	70.1	60.6	51.7	46.6
		MCWB	42.9	41.3	46.1	53.0	58.0	62.8	65.1	63.7	60.1	55.6	48.6	44.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.5	49.0	54.6	60.5	67.5	71.0	71.1	71.1	67.9	62.8	56.7	51.8
		MCDB	52.6	50.1	62.9	72.9	79.5	84.1	83.4	84.4	79.8	69.6	60.0	54.5
	2%	WB	47.5	45.9	51.6	57.9	64.9	68.5	69.6	69.2	64.9	60.0	53.6	49.0
		MCDB	49.6	49.2	58.3	68.3	76.9	79.8	81.7	80.7	74.7	65.9	56.4	50.6
	5%	WB	44.7	43.6	49.2	56.0	62.4	66.6	68.2	67.6	63.2	57.7	51.1	46.8
		MCDB	46.4	46.7	54.4	66.3	72.0	77.5	79.4	78.2	71.7	62.6	53.6	48.8
	10%	WB	42.4	41.1	46.7	54.0	60.2	64.5	66.8	65.9	61.3	55.8	49.2	44.5
		MCDB	44.0	44.0	51.4	63.9	69.0	74.3	77.9	75.0	69.0	59.9	51.0	46.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	10.8	14.6	19.1	19.0	18.8	18.2	18.4	17.4	13.8	9.7	8.3
		MCDBR	9.7	15.4	22.2	24.9	25.4	25.4	25.2	25.2	23.7	18.7	12.4	10.4
		MCWBR	8.3	11.3	13.5	12.8	11.0	10.1	8.9	9.7	10.9	11.1	9.2	8.4
	5% WB	MCDBR	9.2	14.0	19.4	22.7	22.6	22.1	20.9	20.4	20.2	16.8	11.5	10.0
		MCWBR	8.2	10.8	12.8	12.4	10.6	10.1	9.0	9.6	10.6	10.8	9.2	8.6
(40) Clear-Sky Solar Irradiance	taub		0.327	0.353	0.394	0.419	0.416	0.413	0.434	0.428	0.389	0.376	0.351	0.320
	taud		2.310	2.286	2.193	2.198	2.250	2.283	2.236	2.268	2.351	2.347	2.352	2.326
	Ebn at Noon		207	236	248	258	265	267	258	253	249	225	196	188
	Edh at Noon		20	28	36	41	41	40	41	38	31	26	20	17
(44) All-Sky Solar Radiation	RadAvg		203	424	748	1250	1545	1644	1542	1327	967	539	248	151
	RadStd		23	73	85	162	184	130	203	108	111	65	32	16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page